Department of Biotechnology, School of Energy Technology

B.Tech Biotechnology

Course Matrix for Semester – I (Batch 2024-2028)

Sem.	Category Code	Course Code	Course Name	Th	Tut	Pra	Hrs	Cr
I	HSC	24HS101T	English Communication	2	0	0	2	2
	BSC	24MA102T	Mathematics for Biotechnology – I	3	1	0	4	4
	ESC	24CP101T	Computer Programming – I	1	0	0	1	1
	ESC	24CP101P	Computer Programming Laboratory – I	0	0	2	2	1
	BSC	24PH103T	Modern Physics	3	0	0	3	3
	BSC	24PH103P	Modern Physics Laboratory	0	0	2	2	1
	BSC	24CV101T	Environmental Science	2	0	0	2	2
	ESC	24ME101P	Workshop Practices	0	0	2	2	1
	ESC	24ME102P	Engineering Graphics	0	0	4	4	2
	HSC	24HS102T	Universal Human Values	1	0	0	1	1
	BSC	24BT102T	Biology for Engineers	2	0	0	2	2

Syllabus for the Courses of Semester – I (Batch 2024-2028)

24HS101T					English Communication					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

1. Understand of the fundamental elements of communication in English language.
2. Correct expression in the English language at a basic level
3. Using appropriate vocabulary, grammar, effective paragraph construction, writing in day-to-day scenarios, including digital platforms
4. To learn and apply communication skills in different public and interpersonal contexts.

UNIT 1 INTRODUCTION TO COMMUNICATION **07 HRS.**

- The Fundamentals of Language and Communication
- Why English?
- Types of Communication
- Barriers to effective Communication
- Digital Communication

UNIT II: ENGLISH GRAMMAR AND VOCABULARY **07 Hrs.**

- Tenses, Articles, prepositions, active voice passive voice and concord
- Tools for vocabulary building
- Homophones, homonyms, one word substitution, antonyms, synonyms, Root Words, Prefixes and Suffixes, Connotations. Collocations, Idioms. phrases

UNIT III: RECEPTIVE SKILLS **07 Hrs.**

- Listening Skills: Difference between listening and hearing, Active listening and passive listening
- Types of listening, Traits of good listener
- Reading Skills: Why reading is important, Effective reading techniques, Speed Reading, The SQ3R Method

UNIT IV: COMPENDIUM SKILLS **07 Hrs.**

- Note Taking and Note Making: physical, digital, collective
- Summarizing
- Creating e-content

TOTAL: 28 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Demonstrate an understanding of the fundamentals of communication, including the basic components of the communication process and its significance in personal, academic, and professional contexts.

CO2 : Apply grammatical rules accurately in written and spoken communication to enhance clarity, coherence, and precision.

CO3 : Employ effective strategies for vocabulary development, such as word association, context clues, and mnemonic devices.

CO4 : Enhance their ability to identify and interpret implicit meanings, inference, and figurative language in both written and spoken texts, contributing to deeper comprehension and critical analysis.

CO5 : Develop strong written communication skills to effectively convey complex ideas and information in a clear, concise, and coherent manner within the compendium.

CO6 : Apply ethical communication principles and standards of academic integrity when creating and sharing compendiums and avoiding plagiarism.

TEXT/REFERENCE BOOKS

1. Effective Communication Skills. Kul Bhushan Kumar, Khanna Book Publishing, 2022.
2. Remedial English Grammar. F.T. Wood. Macmillan.2007
3. Sharma, Sangeeta and Binod Mishra. Communication Skills for Engineers and Scientists. New Delhi: PHI Learning Pvt. Ltd., 2009.
4. Kaul, Asha. Business Communication. Delhi: Prentice-Hall of India, 2006.

24MA102T					Mathematics for Biotechnology – I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To make students acquainted with the basics of sets, relations, and functions
2. To learn the fundamentals of differential and integral calculus.
3. To understand the use of calculus in real-world applications.
4. To introduce the concept of matrix, determinants, and their use to solve systems of equations.

UNIT I: SET RELATIONS AND FUNCTIONS **10 Hrs.**

Sets and their representation. Union, intersection, and complement. Mapping or function. One-one, onto mappings. Inverse and composite mappings.

UNIT II: BASICS OF DIFFERENTIAL CALCULUS **10 Hrs.**

Limit, continuity, Differentiability of elementary functions, Increasing and Decreasing Functions, Maxima and Minima.

UNIT III: BASICS OF INTEGRAL CALCULUS **10 Hrs.**

Integrals of elementary functions. Substitution and partial fractions. Definite Integral as a limit of sum. Properties of definite integrals. Application to areas and arc lengths.

UNIT IV: MATRICES AND DETERMINANTS **12 Hrs.**

Algebra of matrices. Determinant of a square matrix. Properties of determinants. Some simple types of matrices. The inverse of a matrix. The rank of a matrix, consistency of the system of linear equation upto 3rd order. Eigen values and Eigenvectors of a real matrix, Properties of Eigen values and Eigen vectors, Diagonalization of a matrix.

TOTAL: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand the basic concepts of single variable calculus and matrices.
CO2 : Demonstrate basic matrix operations and determinants.
CO3 : Apply the concepts of calculus to find maxima, minima and area under the curve.
CO4 : Analyse the use of single variable calculus and matrices in biotechnology problems.
CO5 : Evaluate the determinant of different types of matrices, differentiation and integration of functions.
CO6 : Develop basic understanding of matrices and calculus in solving various biotechnology problems.

TEXT/REFERENCE BOOKS

1. B. S. Grewal, Elementary Engineering Mathematics, Khanna Publisher.
2. Seymour Lipschutz, Schaum's Outline of Calculus, Tata McGraw Hill.
5. Nita H. Shah, Foram A. Thakkar, Matrix and Determinant Fundamentals and Applications, CRC Press

24CP101T					Computer Programming – I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
1	0	0	1	1	25	50	25	-	-	100

COURSE OBJECTIVES

1. To understand the usage of operators and data types.
2. To apply different types of Conditional and looping statement.
3. To create different types of data collections.
4. To implement user defined function.
5. To perform different operations upon files.

UNIT 1: Basics of Programming **3 Hrs.**

Introduction to Computer Programming, Features of C language, Structure of C program, program execution flow, C Tokens, variables, Data types, Operators, Decision control statements-if, switch, go to statement. Loop control structures- while, do-while, for loop, Break statement, Continue statement

UNIT 2: Derived Data types **4 Hrs.**

Array: one dimensional and multidimensional array, Declaration, initialization, Array Manipulations. Matrix operations, String-Basic Concepts, Inbuilt String manipulation Functions, Pointer, Pointer arithmetic, Pointer to pointer, Array of Pointers

UNIT 3: Functions and Structures **4 Hrs.**

Introduction to user defined functions, Types of Functions, Call by value-call by reference, recursion, pointers to functions, Structures, Array of Structure, Union

UNIT 4: Files Handling **3 Hrs.**

File handling in C, Different types of files, Operations on Files such as File creation, File deletion, File access modes such as read, write, append, File concatenation, File handling using seek function.

Total: 14 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Understand basics of programming.
CO2: Identify different programming constructs for a problem.
CO3: Apply appropriate derived data type for data storage.
CO4: Prepare a user defined data type based on data attributes.
CO5: Construct user defined functions for problem solving.
CO6: Analyse different data structure based on application requirement.

TEXT/REFERENCE BOOKS

1. Kernighan & Ritchie, C Programming Language, PHI
2. K. N. King , C Programming: A Modern Approach, W.W. Nortorn
3. David Griffiths and Dawn Griffiths, Head First C: A Brain-Friendly Guide, O'Reilly
4. E.Balaguruswamy, Programming in ANSI C, McGraw-Hill
5. Y.P. Kanetkar, Let us C, BPB Publication
6. Y.P. Kanetkar , Pointers in C, BPB Publications

24CP101P					Computer Programming Laboratory – I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To implement basic programming concepts.
2. To create different types of data collections.
3. To implement user defined function.
4. To perform different file handling operations.

List of Experiments:

1. **Introduction to Computer Programming**: Understanding compilation process through a simple C program, program execution flow, C Tokens, variables and keywords and identifiers, types of C constants and variables.

2. **Simple and formatted Input Output Operations:** Data types, Operators, Input /output statements in C, Formatted I/O, format specifiers, escaper sequences
3. **Decision making and branching:** if, if-else, if-else ladder, switch, go to statement, conditional operator statement

4. **Looping control structures:** while, do-while, for loop, Break statement, Continue statement
5. **Derived Data Type: Array and Strings:** One dimensional and multidimensional array, Declaration, initialization, Array Manipulations. Matrix operations, Basic Concepts, Inbuilt String manipulation with and without using inbuilt functions.
6. **Derived Data Type: Structure and Union:** structure, arrays and structures, structures and functions, pointer to structure, typedef, unions
7. **Functions:** Introduction to user defined functions, Types of Functions, Call by value-call by reference, header file creation, recursion, pointers to functions, arrays and functions
8. **Pointers:** Pointer's basics, use of &, * operator in context to pointers, Pointer arithmetic, Array and String processing using pointer, pointer to pointer, Array of Pointers
9. **File Handling in C:** File handling in C, Different types of files, Operations on Files, File handling functions.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Identify the use of appropriate naming conventions and programming style including appropriate comment density.
CO2: Implement a basic C program using appropriate control structure.
CO3: Apply appropriate derived data types based on data attributes.
CO4: Develop solutions with pointers and utilize them to access strings and structures.
CO5: Design user defined functions for problem solving and reuse them across different programs.
CO6: Apply suitable file handling functions and operations.

TEXT/REFERENCE BOOKS

1. Kernighan & Ritchie, C Programming Language, PHI.
2. K. N. King , C Programming: A Modern Approach, W.W. Norton.
3. E. Balaguruswamy, Programming in ANSI C, McGraw-Hill.
4. Y.P. Kanetkar,Let us C, BPB Publication.
5. Y.P. Kanetkar, Pointers in C, BPB Publication.

24PH103T					Modern Physics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. Understanding concepts of modern physics.
2. Explaining the physics of EM waves and its propagation.
3. Applying the Acquired basic knowledge of solid state physics.
4. Analyze the Concepts of nuclear radiation physics.
5. Estimate and correlate the concepts learned so far for various engineering applications.
6. Apply the fundamentals designs of modern physics to solve complex physical problems.

UNIT I: MODERN PHYSICS **09 Hrs.**

Review of quantum concepts: particle nature of light, photoelectric effect, Compton effect, matter waves, wave packets, phase and group velocity, Interference, Diffraction and Polarization, Engineering Physics related applications. Davisson Germer experiment, Heisenberg uncertainty principle.

UNIT II: ELECTROMAGNETIC WAVES **12 Hrs.**

Physical and mathematical concepts of gradient, divergence and curl, Gauss theorem, applications in gravitation and electrostatics. Stokes' theorem and engineering Physics related applications. Equation of continuity, Biot Savart law – Ampere's law – magnetization and magnetic intensity, Faraday's law of Maxwell's equations, wave equation for electromagnetic radiation, applications of optical fibers in communication.

UNIT III: SOLID STATE PHYSICS **12 Hrs.**

Crystalline and Amorphous Solids, Ionic Crystals, Covalent Crystals, Van der Waals Bond, Band Theory of Solids, Semiconductor Devices, Electrical conductivity, Resistivity, Magnetism, Superconductivity, Introduction to BCS Theory.
Concepts of LASER, Interaction of radiation of matter-quantum mechanical view, characteristics and Types of laser, Engineering Physics related application of lasers.

UNIT IV: NUCLEAR RADIATION PHYSICS **09 Hrs.**

Mass defect, binding energy, Radioactivity, Types of Radiation, Interaction of Radiation with matter, Radiation detector, nuclear reactions, elements of nuclear reactors, fission and fusion, Engineering Physics related problems.

TOTAL: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : To relate to shape the engineering perspective in the student mind.
CO2 : Aims to provide an understanding to analyse the physical phenomena of various physics concepts.
CO3 : To develop an analytical perspective in the student.
CO4 : To enable understanding in the students the importance of application of already studied topics.
CO5 : To explain and relate the importance of interdisciplinary problems.
CO6 : To strengthen problem solving attitude in physics using mathematical tools.

TEXT/REFERENCE BOOKS

1. Resnick, Halliday and Krane, Physics part I and II, John Wiely
2. Ghatak, Optics, Tata McGraw Hill
3. Purcell E.M. Electricity and Magnetism - Berkeley Physics Course, Vol.2, Tata McGraw-Hill.
4. Kittel C., Knight W.O. and Ruderman M.A., Mechanics - Berkeley Physics Course, Tata McGraw-Hill.
5. Griffith D.J.H., Introduction to Electrodynamics - Prentice Hall, India.
6. M. N. Avadhanulu, A text book of engineering Physics, S. Chand & Company, Ltd.
7. Feyman R.P., Leighton R.B. and Sands M. The Feyman Lectures on Physics, Vol. 1., Narosa Publication.

24PH103P					Modern Physics Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. Understand the characteristics of waves, semiconductor, heat pump, LASER.
2. Enhance knowledge on application of physics in engineering
3. To develop intellectual communication skills and discuss the basic understanding of various experimental principles involved
4. Demonstrate practical knowledge by applying experimental methods to correlate with the theory.
5. Apply the analytical techniques and graphical analysis to the experimental data

LIST OF EXPERIMENTS

1 Determining Plank's constant and inverse square law
2 Study of Photoconductivity
3 Study of Bio-Savart's Law
4 Determining e/m by Thomson's method
5 Study of Hall Effect.
6 Experiments on single and double slit diffraction and interference with He-Ne Laser
7 Study of I-V characteristics of p-n diode.
8 Determination of thermal conductivity of different solids
9 To measure resistivity of semiconductor by Four Probe method and determination of band gap.
10 Study of Interference using Newton's Ring experiment.
11 To study G.M. tube characteristics and to calculate the dead time.
12 Energy calibration of CsI:Tl radiation detector and energy analysis of an unknown gamma source
13 To determine the numerical aperture of a given fibre optics cable using the far field measurements.
14 Experiments with heat pump
15 Study of Polarization of light using LASER

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Analysis the engineering problems and design the components for the solution
CO2 : Developing skills to utilize the different tools for engineering problems
CO3 : Analyse the results and correlate with theory and its application in industries
CO4 : Design the set-up and utilize for component analysis
CO5 : Identifying the problem and creating the solutions for research and development
CO6 : Analyse the scientific data and learn to be efficient as individual and a team member

TEXT/REFERENCE BOOKS

1. W.R. Runyan , Semiconductor Measurements and Instrumentation, McGraw Hill.
2. Sayer M. & Mansingh A., Measurement, Instrumentation & Experiment Design in Physics and Engineering, Prentice Hall India.
3. Melissinos A.C. and Napolitano J, Experiments in Modern Physics, Academic Press.
4. Nakra B.C. &. Chaudhery K.K , Instrumentation Measurements & Analysis, Tata McGraw Hill.
5. ORTEC Lab Manual, Experiments in Nuclear Science, ORTEC.

24CV101T					Environmental Science					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

1. To develop a comprehensive perspective of environment and sustainable development
2. To understand the causes and effects of various types of pollution
3. To develop an understanding of the various strategies for controlling the pollution
4. To introduce the emerging environmental domains

UNIT I : INTRODUCTION TO ENVIRONMENT **05 Hrs.**

Sustainable Development; Sustainable Development Goals; Environmental Studies – Its importance and Multidisciplinary nature, Introduction to Environmental Parameters and their standards (air, water, soil, noise, etc.); Ecosystem and its types, Ideal ecosystem, Biodiversity : Its importance and conservation.

UNIT II : MULTI-SCALE ENVIRONMENTAL POLLUTION (GLOBAL, REGIONAL AND LOCAL) **06 Hrs.**

Pollution, Causes and Effects of different types of pollution : Air Pollution, Water Pollution, Soil Pollution, Solid Waste (organic and Inorganic) Pollution, Hazardous Waste Pollution, Marine Pollution, Noise Pollution, Thermal Pollution, Radioactive Pollution; Introduction to man-made disasters like floods, heat waves, landslides, etc., Introduction to the various instruments for measuring air pollution, water pollution, noise, etc.

UNIT III : ENVIRONMENTAL POLLUTION CONTROL STRATEGIES **09 Hrs.**

Multi-approaches for reducing various types of pollution: Introduction to Water and Wastewater treatment technologies, Air and Noise pollution control techniques, Introduction to different environmental management concepts like Swachh Bharat Mission, Mission LiFE (Lifestyle For Environment), etc. Indian Culture and Traditional Wisdom for managing environment

UNIT IV: EMERGING ENVIRONMENTAL MANAGEMENTS DOMAINS **08 Hrs.**

Concept of Zero Liquid Discharge (ZLD) and the reuse of the treated wastewater, Green Credit Rules - 2023, Clean Development Mechanisms (CDM) and Carbon Credits, Green Buildings, Carbon Footprint and Water Footprint, Green Business, International Environmental Laws, Environmental Auditing

TOTAL: 28 Hrs.

COURSE OUTCOMES:

On completion of the course, student will be able to:

CO-1: Demonstrate comprehension of sustainable development and environmental aspects.
CO-2: Recognize the interdisciplinary characteristics inherent in Environmental studies.
CO-3: Evaluate the impact of various pollutants on the environment.
CO-4: Assess the efficacy of different technologies for environmental pollution control.
CO-5: Analyze different environmental management policies and their implications.
CO-6: Synthesize knowledge about emerging environmental management paradigms.

TEXT-BOOK AND REFERENCE BOOKS:

1. Bharucha Erach, Textbook for Environmental Studies, UGC New Delhi.
2. Daniel B. Botkin & Edwards A. Keller, Environmental Science, Wiley India edition.
3. Miller T. G. Jr., 2006. Environmental Science, Clengage Learning.
4. R. Rajagopalan, Environmental Studies, Oxford University Press.
5. Gilbert Masters and Wendell P. Ela, Introduction to Environmental Engineering and Science, PHI.

24ME101P					Workshop Practices					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To understand safety in various manufacturing processes.
2. Learn how to use various measuring tools for engineering applications.
3. Hands on training and preparation of job using wood, metal and sheet as per drawing.
4. Understand various manufacturing processes like machining, welding, soldering and 3D printing for prototypes.

LIST OF EXPERIMENTS

1 Introduction to Workshop safety, layout and identification of various materials- plastic, wood, metals-ferrous and nonferrous, rubber, glass etc.
2 Use of measuring tools for engineering applications
3 Fitting job: Detailed drawing of work piece, use of fitting tools and job preparation.
4 Carpentry job: Detailed drawing of work piece, use of carpentry tools and job preparation.
5 Sheet metal job: Detailed drawing of work piece, use of sheet metal working tools and job preparation.
6 Plumbing job: Internal/External threading, piping network using Tees, Elbows, Reducer, Bends etc
7 Hands on training on mini lathe and milling machine
8 Demonstration of welding, brazing and soldering
9 Soldering and desoldering for PCB
10 3D printing using polymer and metal.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Define workshop safety and various engineering materials
CO2 : Understand various measuring equipment
CO3 : Apply various workshop tools in preparing job for carpentry, fitting, sheet metal and plumbing
CO4 : Examine various manufacturing operations like welding and machining
CO5 : Evaluate soldering operation for PCB
CO6 : Create prototype using 3D printing

TEXT/REFERENCE BOOKS

1. S. K. Hajra Choudhury, Elements of Workshop Technology, Vol. I & II, Media Promoters and Publishers.
2. H. S. Bawa, Workshop Practice, Tata-McGraw Hill.
3. Kalpakjian S. And Steven S. Schmid, Manufacturing Engineering and Technology, Pearson Education India Edition.
4. Rao P.N., Manufacturing Technology, Vol. I and Vol. II, Tata McGraw Hill House.

24ME102P					Engineering Graphics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
0	0	4	2	4	--	--	--	50	50	100

COURSE OBJECTIVES

1. To cover the fundamental of engineering drawing and standards used in drawing.
2. To explain the students to communicate ideas using orthographic and isometric projection methods.
3. To help students to use CAD software to prepare drawings.
4. To demonstrate the presentation of drawing using sketching and 3D modelling in CAD tool.

LIST OF EXPERIMENTS

1 Introduction to Engineering Graphics: Principles of engineering graphics and their significance, drawing instruments & accessories, lettering and numbering, types of lines, dimensioning methods, basic geometric drawing, reading a drawing.

2 Orthographic Projection: Introduction to projection, types of projection, 1st angle and 3rd angle projection

3 Isometric Projection: Principles of isometric projection – isometric scale, isometric views, conventions, conversion of isometric views to orthographic views and vice-versa

4 Projection of Solids and Development of Surface: Classification of solids, projections of solids like cylinder, cone, pyramid, and prism with its inclination to reference plane, development of surfaces of right regular solids - prism, pyramid, cylinder and cone.

5 Introduction of Computer Aided Engineering Drawing: Demonstrating knowledge of the theory of CAD software, use of software in drawing, CAD software user interface, commands, Coordinate System, menus and toolbars, planes, dimensioning, saving of files, Select and erase objects, zoom tools, and others

6 Basic sketching using CAD tool: Sketch entities using tools – origin, points, lines, circle, arcs, polygons, fillets and chamfer, trim, extend and offset, copy, cut, delete and others

7 Advanced sketching using CAD tool: Sketching entities using relation constrains, Mirror, Patterning, full definition of drawing and others

8 Basic 3D modelling using CAD tool: Extrude, cut, drawing on different planes, editing, symmetric, revolving, and others

9 Computer aided drawing sheets: Preparing drawing sheets, creating different views, section view, drawing templates, and others

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recalling the fundamentals of engineering graphics by considering basic rules of drawing, dimensioning, and labelling.

CO2 : Explain the principle of projection using orthographic and isometric projection.

CO3 : Represent the 2-dimensional drawing using CAD tool.

CO4 : Construct the 3-dimensional geometries using CAD tool.

CO5 : Apply the concept of engineering drawing by organizing drawing views and applying necessary dimensions by preparing drawing sheets

CO6 : Analyse the intricate details of solid using projection of solid, sectioning of solid and development of lateral surfaces.

TEXT/REFERENCE BOOKS

1. Bhatt N.D., Panchal V.M. & Ingle P.R., Engineering Drawing, Charotar Publishing.
2. Shah P.J., Engineering Graphics, S. Chand Publishing.
3. Agrawal, B. & Agrawal C. M., Engineering Drawing, Tata McGraw Hill Publishers.
4. Hanifan R, Perfecting Engineering and Technical Drawing, Springer International Publishing Switzerland.
5. Corresponding Set of CAD Software Theory and User Manuals.

24HS102T					Universal Human Values					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
1	0	0	1	1	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the need of nurturing human values through the process in value-based education system.
2. To understand and develop a holistic perspective on self-exploration and being in harmony with family, society and nature.
3. To facilitate the students in understanding harmony at all the levels and applying in their profession and work place to lead an ethical life.

UNIT I: HUMAN VALUES AND PROCESS OF VALUE EDUCATION **04 Hrs.**

Human values, human aspirations and the ultimate goal, understanding happiness and prosperity, appraise the meaning of satisfaction and happiness in current scenario, harmony and compatibility, values imbibed education system and process

UNIT II: KNOWING SELF - HARMONY WITH SELF **04 Hrs.**

Understanding self, capabilities and challenges, understanding material (physical facilities) and spiritual needs - need of mind and body, understanding body as an instrument, harmony between mind and body, synchronizing physical health and mental health, practicing healthy habits for healthier me

UNIT III: HARMONY IN RELATIONSHIP – FAMILY, SOCIETY AND NATURE **03 Hrs.**

Harmony in relationships, values for harmony in any human-human interaction, harmony in family, and society, trust and respect for others, self esteem and ego, equality, equity, inclusion and liberation, concept of '*Vasudhaiva Kutumbakam',* understanding co-existence and sync with nature

UNIT IV: HARMONY IN PROFESSION AND ETHICAL BEHAVIOR **03 Hrs.**

Ethical human conduct, acceptance and respect, appraising the qualities of others, professional competence for enabling harmony in system and enabling universal human order, scope of eco-friendly systems, strategies to reach the harmonious ecosystem to reach Universal Human Order '*Sarvabhauma Vyavastha*'

TOTAL : 14 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

- CO1 - Understand the significance of human values, its need, and process of value education.
- CO2 - Appraise the meaning of happiness and prosperity as short- and long-term goal of life. Understand them and in context of the current scenario
- CO3 - Distinguish between the mind and body, physical and spiritual wellbeing for harmony within self
- CO4 - Assess the value of harmonious relationship based on trust, respect and enduring its role in all human-human relationships to build harmonious society
- CO5 - Understand the importance of harmony with nature and appreciate co-existence for harmonious ecosystem.
- CO6 - Create the perfect professional place and work environment following the ethical practices and strategize to uphold the human values at all the levels and interactions.

TEXT/REFERENCE BOOKS

1. R. R. Gaur, R Sangal, G P Bagaria, A foundation course in Human Values and Professional Ethics, Excel books.
2. A. Nagraj, Jeevan Vidya ek Parichay, Divya Path Sansthan, Amarkantak.
3. A. N. Tripathi, Human Values, New Age Intl. Publishers.
4. M. K. Gandhi. The Story of My Experiments with Truth, Fingerprint Publishing.
5. Ivan Illich, Energy & Equity, The Trinity Press, Worcester, and Harper Collins.
6. E. F. Schumacher, Small is Beautiful: a study of economics as if people mattered, Blond & Briggs, Britain.
7. Sussan George, How the Other Half Dies, Penguin Press.

24BT102T					Biology For Engineers					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the basic biological concepts and their engineering applications.
2. To introduce the students with an understanding of biodesign principles to create novel devices and structures.
3. To provide the students an appreciation of how biological systems can be re-designed as substitute products for natural systems.
4. To study the development of interdisciplinary vision of biological engineering.

UNIT I: INTRODUCTION TO BIOMOLECULES **7 Hrs.**

Cell as the basic unit of life, Overview of cell structure and functions. Carbohydrates: Properties and biological functions. Nucleic
Acids: Structure and roles of DNA and RNA. Proteins: Structure, and its functions, Significance of lipids. Role of biomolecules: Characteristics and roles of enzymes, vitamins, and hormones.

UNIT II: NATURE-BIOINSPIRED MATERIALS AND MECHANISMS (QUALITATIVE) **7 Hrs.**

Echolocation (ultrasonography, sonars), Photosynthesis (photovoltaic cells, bionic leaf). Bird flying (GPS and aircrafts), Lotus leaf
effect (Super hydrophobic and self-cleaning surfaces), Plant burrs (Velcro), Shark skin (Friction reducing swim suits), Kingfisher beak (Bullet train). Human Blood substitutes- hemoglobin-based oxygen carriers (HBOCs) and perflourocarbons (PFCs).

UNIT III: ENGINEERING PERSPECTIVES OF BIOLOGICAL SCIENCES **7 Hrs.**

Biology and engineering crosstalk –Hybridoma technology, Plant Tissue Culture, Animal Tissue Culture; Tissue Engineering,
Introduction to Biomimetics and Biomimicry, Nanobiotechnology, Introduction to Radiology, High-throughput diagnostics in clinics, Bioprocessing and Biomaterials.

UNIT IV: TRENDS IN BIOENGINEERING **7 Hrs.**

Stem cell and its applications, Bio printing techniques and materials. Applications of Bioinformatics, Artificial Intelligence for
diagnosis and detection of communicable and non-communicable diseases, Biosensors in healthcare, Environmental monitoring.

TOTAL: 28 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Gain fundamental knowledge of origin of life.
CO2 : Explain the functions of cell and its components.
CO3 : Evaluate the principles of design and development, for exploring novel bioengineering projects.
CO4 : Compare biology with engineering
CO5 : Corroborate the concepts of biomimetic for specific requirements
CO6 : Think critically towards exploring innovative biobased solutions for socially relevant problems.

TEXT/REFERENCE BOOKS

1. Thyagarajan S., Selvamurugan N., Rajesh M.P., Nazeer R.A., Thilagaraj W., Barathi S., and Jaganthan M., **"Biology for Engineers**", Tata McGraw-Hill, New Delhi, 2012.
2. Arthur T. Johnson, **"Biology for Engineers"**, CRC Press, Taylor and Francis, 2011
3. Geoffrey M.Cooper**, "The Cell: A molecular approach"**, ASM Press, 2007.
4. Sohini Singh and Tanu Allen, **"Biology for Engineers"**, Vayu Education of India, New Delhi, 2014.
5. Yoseph Bar-Cohen, **"Biomimetics: Nature-Based Innovation**" , 1st edition, 2012, CRC Pres